

REPORT

WHO INFLUENZA CENTRE

LONDON

March 2008

**CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2, AND B
VIRUSES ISOLATED SEPTEMBER 2007 TO FEBRUARY 2008**

**WHO Collaborating Centre for
Reference and Research on Influenza
National Institute for Medical Research
The Ridgeway
Mill Hill
London, NW7 1AA**

**Dr. Alan J. Hay (Director)
Dr. Rodney Daniels (Deputy Director)
Dr. Yi Pu Lin (Assistant Director)
Dr. Xiang Zheng
Miss Victoria Gregory
Mr. Michael Bennett
Ms Lynne Whittaker**

Tel: 020 8959 3666; 020 8816 2152

Fax: 020 8906 4477

Email: ahay@nimr.mrc.ac.uk; whocc@nimr.mrc.ac.uk

Influenza activity, October 2007 to March 2008

Influenza activity was generally low in most parts of the world. Activity increased in Asia and North America during November to January and in Europe during December and January. During the early part of the season, influenza AH1N1 viruses predominated in most parts of the northern hemisphere. In the United States AH3N2 viruses increased to become predominant during February. While the proportion of H3N2 viruses remained low in Europe, the proportion of influenza B viruses increased steadily during 2008.

Antigenic and genetic characteristics of influenza viruses isolated October 2007 to February 2008

Approximately 700 influenza viruses, isolated in 30 countries during October to February, were characterized at the WHO Influenza Centre. The majority were AH1N1 (77%), and only a few (2%) were of the H3N2 subtype. Influenza B viruses accounted for 21% of isolates, of which the majority (86%) were of the B/Yamagata lineage (Table 1).

AH1N1 viruses

Although many of the H1N1 viruses were shown in HI tests to be antigenically closely related to the current vaccine strain A/Solomon Islands/3/2006, approximately half were distinguishable and gave patterns of reactivity in the HI tests more closely related to those obtained with more recent reference viruses, including A/Netherlands/345/07, A/Egypt/10/07 and/or the recently recommended vaccine virus A/Brisbane/59/07 (Table 2). Very few were A/New Caledonia/20/99-like and these were mainly among viruses from Madagascar, accounting for 18 % of those isolates.

With few exceptions, the **HA and NA sequences** of European and African viruses fell within clade 2B, represented by A/Brisbane/59/07 (Figures 1 and 3). The HA sequences differed by 6 common amino acids and the NA sequences by 14 amino acids from the sequence of A/Solomon Islands/3/06 (Figures 2 and 4, Table 3). HA sequences of some of the Austrian viruses, like those of many Japanese isolates, fell within clade 2C, represented by A/Hong Kong/2652/06. All the clade 2 HA sequences possessed the K140E substitution, of prime importance in determining their antigenic characteristics. The HA and NA sequences of A/New Caledonia/20/99-like Madagascar isolates fell within clade 1, represented by A/Thessaloniki/24/05 (Figures 1 and 3). In general, there has been relatively little change in these HA sequences and the corresponding NA sequences of equivalent viruses isolated earlier during 2007.

Oseltamivir resistance

A substantial number of viruses (approx.7% of total, Table 1) especially those isolated in Europe, were shown to possess high level resistance to oseltamivir, either by enzyme inhibition or the presence of the H275Y mutation (H274Y in N2 numbering) in the NA, or both. These results were obtained in collaboration with M. Zambon and A. Lackenby, under the umbrella of the EU funded VirGil network on antiviral drug resistance, as well

as from data supplied by the labs submitting the viruses. Table 4 lists the data recently available for different European countries (total of 21% resistance to date) and shows that the proportion of resistant viruses varies markedly between countries, from 69% for Norway to 1% for Italy, with no evidence yet of resistant-viruses in several countries.

The nucleotide sequences of the NAs of many of the resistant viruses, including some from the USA, Australia and Canada, group together and are characterized by the loss of the G354D change, typical of earlier NAs of clade 2B, in addition to the H275Y change, while others from the UK, as well as Japan, Thailand and the USA, are distributed more generally within the subclade (Figure 3). It is apparent, therefore, that the emergence of some of the oseltamivir-resistant viruses occurred independently. Figure 5 shows the locations of amino acid substitutions which distinguish clade 2B from other NAs; of these only the G249K substitution occurs in close proximity to residue 275, in relation to possible complementation between the changes.

The HA gene sequences of resistant and sensitive viruses did not group separately in the phylogenetic tree (Figure 1). Furthermore, there were no common amino acid differences between the HAs of resistant and sensitive viruses and they were not distinguished in HI tests (Table 2).

The majority of the oseltamivir-resistant viruses tested were shown to retain sensitivity to zanamivir and amantadine/rimantadine.

Amantadine resistance

M2 protein gene sequences obtained for viruses with clade 1 or 2B HAs did not contain amantadine resistance mutations. As previously, viruses with HAs which fell within clade 2 C, represented by A/HK/2652/06, possessed the resistance mutation S31N, indicative of resistance to amantadine and rimantadine.

AH3N2 viruses

Although some of the H3N2 viruses were shown in HI tests to be antigenically closely related to the A/Wisconsin/67/05 and A/Brisbane/10/07 vaccine viruses, most gave low HI titres, relative to the homologous virus, with antisera to the egg-grown vaccine strains and more recent reference viruses (Table 5). MDCK cell-grown reference viruses, such as A/Trieste/25c/07, also gave relatively low homologous HI titres with their corresponding ferret antisera, compared with egg-passaged reference viruses. However, ferret antisera to MDCK cell-grown viruses gave similar patterns of reactivity with the panel of reference viruses as antisera to egg-grown viruses. Of the relatively few H3 viruses received, most gave patterns of HI reactivity similar to that of A/Trieste/25c/07 (Table 5).

The **HA sequences** of recent isolates fell within the clade represented by the HA of A/Brisbane/10/07 and characterized by the amino acid changes, relative to the HA of A/Wisconsin/67/05, G50E and K140I (Figures 6 and 7). The **NA sequences** of these viruses were shown to fall into the corresponding clade (Figures 8 and 9). The sequences were similar to those of equivalent viruses circulating earlier in 2007 and there was no evidence for the emergence of a significant distinct subclade (Table 6).

Amantadine resistance

M2 sequences of selected viruses showed that, as for earlier isolates, those with HAs which fell within the “Brisbane/10/07” clade possessed the amantadine resistance mutation S31N, indicating retention of amantadine resistance (Figure 6).

B viruses

The majority of B viruses received were of the B/Yamagata lineage. However, the relative proportions of B/Victoria and B/Yamagata lineage viruses isolated in different countries varied; for example, B/Victoria lineage viruses were more prominent among B viruses isolated in Hong Kong and Italy.

Many of the **B/Yamagata lineage viruses** were antigenically closely related to the vaccine strains, B/Florida/4/06 and B/Brisbane/3/07, and/or to the reference viruses B/Florida/7/04 and B/Egypt/144/05 (Table 7). These 4 reference viruses were not distinguishable in the HI tests. A number of viruses gave low HI titres with antisera to these viruses; however, there was no consistent correlation between differences in patterns of HI reactivity and differences in HA sequence.

HA sequences of recent B/Yamagata lineage isolates were similar to those of viruses isolated earlier during 2007 and fell into 3 principal phylogenetic subgroups, two of which are represented by the reference/vaccine viruses B/Florida/4/06 and B/Brisbane/3/07 (and B/Sendai-H/114/07) (Figure 10). Amino acid differences in **HA** and **NA** (Figures 10-13) characteristic of these sequence variants are listed in Table 8.

Most of the few **B/Victoria lineage viruses** were antigenically closely related to B/Hong Kong/45/05, but several gave reduced HI titres with antiserum to the egg-grown vaccine strain B/Malaysia/2506/04 and more recent reference virus B/Victoria/304/06, as previously observed (Table 9).

Their **HA sequences** were close to those of B/Victoria lineage viruses isolated earlier during 2007 and B/Malaysia/2506/04 (Figures 14 and 15). Most of the **NA sequences** were similar to those obtained for viruses isolated during 2006 and 2007 and differed from that of B/Malaysia/2506/04 by 8 amino acids (Figures 12 and 13; Table 8).

Acknowledgements

We thank all who have contributed information and viruses, and associated data, to the WHO Global Influenza Surveillance Network, which provide the basis for our current understanding of recently circulating influenza viruses, and this brief report. In addition to data shared among the other 3 Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Austria, Denmark, France, Germany, Italy, Norway, Portugal, Sweden, and UK.

Table 1. Human influenza isolates, October 2007 - February 2008

Country	H1N1		H3N2	B	
	A/NC/20/99 ¹	A/SI/3/06 ²		Yamagata lineage	Victoria lineage
October					
Egypt		1			
France	1	2		1	2
Madagascar	5	18		1	
Netherlands		1	1	1	
Norway				1	
Senegal		9			
Slovakia		1			
Spain		3	1		
Sweden		1			
United Kingdom				1	
November					
Algeria		2			
Egypt		9			
France		5 (1)	4	2	
Germany		1			
Hong Kong		1			
Italy		1			
Latvia		6			
Madagascar	3	24		2	
Norway		8 (5)		3	
Slovakia		2			
Slovenia		2			
Spain		6			
Sweden		2		2	
Switzerland		1	1		1
December					
Austria		6		1	
Belgium		2	2	1	
Czech Republic		1			
Denmark		1		1	
Egypt					1
Finland		2		1	
France		26 (4)	2	3	
Germany		3		4	
Hong Kong		1	2	2	3
Hungary		5			
Ireland				2	
Italy		20		3	5
Latvia		1			
Luxembourg		1			
Madagascar		3			
Norway		13 (12)		4	
Portugal		1			
Slovakia		3			
Spain		24		1	
Sweden		5 (1)		1	
Switzerland		7		1	
United Kingdom		6		2	
January					
Belgium		4 (1)			
Bulgaria		9			
Croatia		6		4	
Czech Republic		23		5	
Denmark		3		3	1
Finland			2		
Germany		8		8	
Greece		25 (2)			
Hungary		7			
Ireland		5			
Italy		56		30	4
Latvia		10			
Luxembourg		11 (1)			
Norway		9 (5)		1	
Portugal		12 (2)		7	
Romania		9		3	
Slovenia		30			2
Spain		20		5	
Sweden		4	1		
Switzerland		2		1	
United Kingdom		1			
February					
Italy				4	1
Spain				4	
Portugal				1	
Russia		9	1	1	
Total = 664	9	500 (34)	17 (2%)	118 (21%)	20

() oseltamivir resistant; 1. A/NC/20/99 includes viruses that are A/New Caledonia/20/99-like and have HAs that may fall within clades 1 or 2; 2. A/SI/3/06 includes viruses with HAs that fall within clade 2 and are antigenically closely related to A/Solomon Islands/3/06 or A/Brisbane/59/07.

Table 2. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infecton ferret sera							
			A/NC	A/Thess	A/HK	A/Fuk	A/SI	A/Neth	A/Eg	A/Bris
			20/99	24/05	2652/06	141/06	3/06	345/07	10/07	59/07
			F18/06	F10/05	F14/06	NIB 1/07	F8/07	F23/07	F4/07	F2/07
REFERENCE VIRUSES										
A/New Caledonia/20/99	09/06/99	Ex	320	640	40	80	40	160	80	160
A/Thessaloniki/24/05	03/02/05	E2 \3	320	1280	40	80	40	160	80	160
A/Hong Kong/2652/2006	17/07/06	E2 \2	40	40	320	320	320	160	320	640
A/Fukushima/141/2006	07/02/06	E4 \2	80	40	160	640	640	160	640	1280
A/Solomon Islands/3/2006	21/08/06	E3 \2	80	160	320	320	640	640	160	640
A/Netherlands/345/2007	12/10/07	MDCK2 \4	80	160	160	160	320	1280	160	640
A/Egypt/10/2007	30/10/07	E1 \1	40	40	160	320	320	160	640	1280
A/Brisbane/59/2007	01/07/07	E2 \1	40	40	160	320	320	160	640	1280
A/Brisbane/59/2007	01/07/07	MDCKx+2 \1	<	<	40	<	<	640	160	320
TEST VIRUSES										
A/Madagascar/2121/2007	31/10/07	MDCK1 \1	160	2560	80	80	80	160	80	160
A/Algeria/G258/2007	12/11/07	C1 \1	40	80	80	40	80	640	320	640
A/Paris/0577/2007 #	Dec-07	MDCKx \1	40	40	80	80	80	320	320	320
A/Bordeaux/1314/2007	04/12/07	MDCK3 \1	40	<	80	40	40	160	160	320
A/Norway/12/2008 #	13/12/07	MDCK1 \1	40	80	80	40	80	320	160	640
A/Firenze/25/2007	20/12/07	MDCK \1	40	40	80	160	160	160	640	160
A/Finland/4/2007	23/12/07	MDCK2 \1	40	80	80	80	160	320	160	640
A/Hungary/3/2007	25/12/07	E3 \1	<	<	160	160	320	80	160	40
A/Zagreb/1982/2008	Jan-08	Ex \1	<	40	80	320	320	160	640	1280
A/Ireland/v482/2008	Jan-08	Cx \1	<	160	160	80	160	1280	160	320
A/Luxembourg/116/2008#	Jan-08	MDCKx \1	80	160	320	320	320	1280	320	640
A/Stockholm/1/2008	Jan-08	MDCK1 \1	<	<	40	40	40	80	80	80
A/Latvia/1-54/2008	01/01/08	MDCK1 \2	40	160	160	320	320	1280	320	640
A/Lisbon/3/2008 #	02/01/08	MDCK3 \1	40	160	160	160	320	1280	320	1280
A/Belgium/G235/2008	02/01/08	MDCKx \3	40	160	160	160	320	1280	160	320
A/Barcelona/00083/2008	03/01/08	MDCK0 \1	<	<	40	<	<	320	160	640
A/Denmark/4/2008	04/01/08	MDCK2 \1	40	80	80	80	80	640	160	640
A/Baden-Wurtemberg/2/2008	07/01/08	MDCKx \1	80	160	40	40	<	160	40	160
A/Hungary/2/2008	07/01/08	C2 \1	40	40	160	320	320	320	640	1280
A/Athens/41/2008#	08/01/08	MDCK1 \1	40	160	160	80	320	1280	—	320
A/Trieste/4/2008	08/01/08	MDCKsiat1 \1	40	160	160	160	320	1280	320	640
A/Lisbon/11/2008	11/01/08	MDCK2 \1	40	160	160	160	160	1280	320	640
A/Bulgaria/062/2008	15/01/08	E1 \1	40	80	160	160	320	320	80	320
A/Volos/108/2008#	15/01/08	MDCK1 \1	40	160	160	80	160	1280	—	320
A/Slovenia/137/2008	10/01/08	MDCK1 \2	40	320	320	160	320	1280	160	320
A/Bucuresti/134/2008	14/01/08	E2 \1	<	<	160	320	160	40	160	320
A/Praha/78/2008	17/01/08	AT1-MDCK2 \1	40	320	160	160	320	1280	320	320
A/Roma-ISS/1/2008	19/01/08	II MDCK \2	40	160	160	160	320	640	320	320
A/Thessalonika/17/2008	23/01/08	MDCK2 \1	80	160	160	160	320	1280	320	640
A/Parma/20/2008	31/01/08	MDCK \1	40	40	80	160	160	160	320	320
A/Perugia/15/2008	31/01/08	I MDCK \1	40	160	160	160	320	1280	320	640
A/St.Petersburg/5/2008	13/02/08	E2 \1	<	40	320	320	320	320	640	1280
A/Kaliningrad/5/2008	13/02/08	C1 \1	<	40	80	<	40	640	640	640

1. < : <40; #, oseltamivir resistant

New vaccine strain

Figure 1. Phylogenetic comparison of H1 HA genes

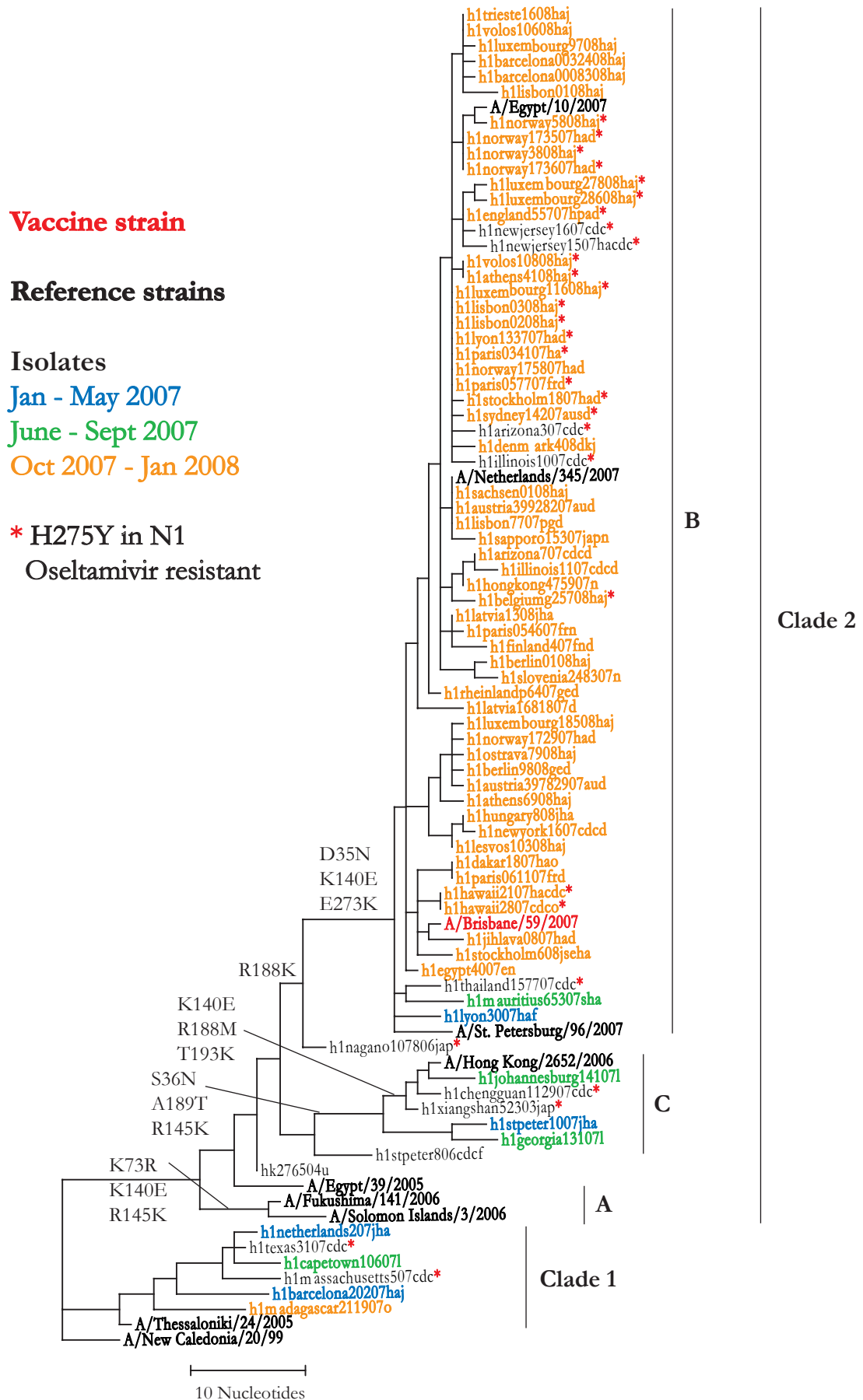


Figure 2. HA1 sequences of H1N1 viruses

	1									100
A/New Caledonia/20/99				d					t	y
A/Thessaloniki/24/2005				d					t	y
A/Egypt/39/2005				d						y
A/Solomon Islands/3/2006				d				r		
A/Fukushima/141/2006				d				r		
A/Hong Kong/2652/2006				dn						
A/Brisbane/59/2007										
A/St. Petersburg/96/2007										
A/Egypt/10/2007										
A/Netherlands/345/2007										
A/Hungary/8/2008										
A/Barcelona/00083/2008										
A/Paris/0546/2007	l									
A/Berlin/98/2008										
A/Norway/1758/2007										
A/Belgium/G257/2008										
A/Paris/0341/2007										
A/Volos/108/2008										
A/Lisbon/03/2008										
A/Norway/1736/2007										
Consensus	DTICIGYHAN	NSTDTVDTVL	EKNVTVTHSV	NLLENSHNGK	LCLLKGIAPL	QLGNCSVAGW	ILGNPECELL	ISKESWSYIV	EKPNPENGTC	YPGHFADYEE
	100									200
A/New Caledonia/20/99				k			v		r	
A/Thessaloniki/24/2005				k					r	
A/Egypt/39/2005				k	k				n_r	
A/Solomon Islands/3/2006			t		k	r			r	k
A/Fukushima/141/2006			t		k				r	
A/Hong Kong/2652/2006					k				mt	k
A/Brisbane/59/2007									i	
A/St. Petersburg/96/2007										
A/Egypt/10/2007									n	
A/Netherlands/345/2007										
A/Hungary/8/2008									n	
A/Barcelona/00083/2008										
A/Paris/0546/2007										
A/Berlin/98/2008		k								
A/Norway/1758/2007									v	
A/Belgium/G257/2008									v	
A/Paris/0341/2007										
A/Volos/108/2008										
A/Lisbon/03/2008										
A/Norway/1736/2007										
Consensus	LREQLVSSVSS	FERFEIFPKE	SSWPNHTVTG	VSASCSHNGE	SSFYRNLLWL	TGKNGLYPNL	SKSYANNKEK	EVLVLWGVHH	PPNIGDQKAL	YHTENAYVSV

Figure 2 continued

	200									300
A/New Caledonia/20/99	r					w	t	e		
Thessaloniki/24/2005	r					f	t	e		
A/Egypt/39/2005								e		
A/Solomon Islands/3/2006								e		
A/Fukushima/141/2006								e		
A/Hong Kong/2652/2006								e		
A/Brisbane/59/2007										
A/St. Petersburg/96/2007			k						l	
A/Egypt/10/2007										
A/Netherlands/345/2007										
A/Hungary/8/2008										
A/Barcelona/00083/2008										
A/Paris/0546/2007				r						
A/Berlin/98/2008										
A/Norway/1758/2007										
A/Belgium/G257/2008										
A/Paris/0341/2007										
A/Volos/108/2008										
A/Lisbon/03/2008										
A/Norway/1736/2007										
Consensus	VSSHYSRKFT	PEIAKRPKVR	DQEGRINYYW	TLLEPGDTII	FEANGNLIAP	RYAFALSRGF	GSGIINSNAP	MDKCDAKCQT	PQGAINSSLP	FQNVHPVTIG
	300		326							
A/New Caledonia/20/99										
A/Thessaloniki/24/2005		a								
A/Egypt/39/2005										
A/Solomon Islands/3/2006										
A/Fukushima/141/2006										
A/Hong Kong/2652/2006										
A/Brisbane/59/2007										
A/St. Petersburg/96/2007										
A/Egypt/10/2007										
A/Netherlands/345/2007										
A/Hungary/8/2008										
A/Barcelona/00083/2008										
A/Paris/0546/2007										
A/Berlin/98/2008										
A/Norway/1758/2007										
A/Belgium/G257/2008										
A/Paris/0341/2007										
A/Volos/108/2008										
A/Lisbon/03/2008										
A/Norway/1736/2007										
Consensus	ECPKYVRS AK	LRMVTGLRNI	PSIQSR							

Figure 3. Phylogenetic comparison of N1 neuraminidase genes

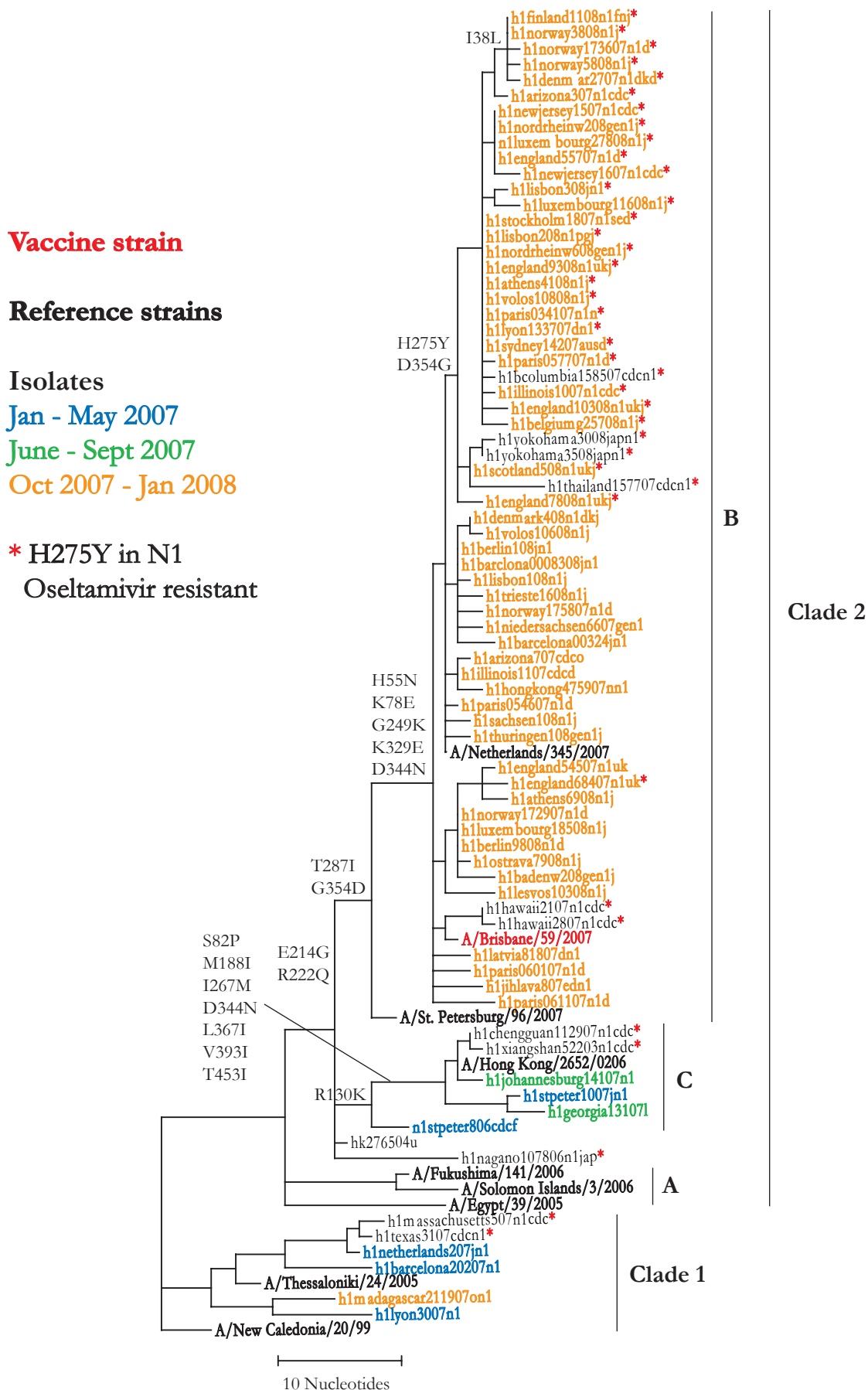


Figure 4. Neuraminidase sequences of H1N1 viruses

	1										100
A/New Caledonia/20/99					h	v				k	
A/Thessaloniki/24/2005					h					k	l
A/Netherlands/2/2007					h			s		k	
A/Massachusetts/5/2007					h			s		k	
A/Egypt/39/2005		t			h		n			k	
A/Solomon Islands/3/2006			i		h			n		ek	
A/Fukushima/141/2006			i		h			n		ek	
A/Hong Kong/2652/2006					h					k	p
A/St. Petersburg/96/2007					h			s		k	
A/Brisbane/59/2007											
A/Barcelona/00083/2008											
A/Paris/0546/2007											
A/Berlin/98/2008											
A/Norway/1758/2007											
A/Belgium/G257/2008			l								
A/Paris/0341/2007											
A/Volos/108/2008											
A/Lisbon/3/2008											
A/Norway/1736/2007						l					
Consensus	MNPNQKIITI	GSISIAIGII	SLMLQIGNII	SIWASHSIQT	GSQNNTGICN	QRIITYENST	WVNHTYVNIN	NTNVVAGEDK	TSVTLAGNSS	LCSISGWAII	
	100										200
A/New Caledonia/20/99											
A/Thessaloniki/24/2005											
A/Netherlands/2/2007											
A/Massachusetts/5/2007											
A/Egypt/39/2005		t									
A/Solomon Islands/3/2006						t		r			
A/Fukushima/141/2006								r			
A/Hong Kong/2652/2006			k						i		
A/St. Petersburg/96/2007											
A/Brisbane/59/2007					g						
A/Barcelona/00083/2008											
A/Paris/0546/2007											
A/Berlin/98/2008											
A/Norway/1758/2007											
A/Belgium/G257/2008											
A/Paris/0341/2007											
A/Volos/108/2008											
A/Lisbon/3/2008											
A/Norway/1736/2007											
Consensus	TKDNSIRIGS	KGDVFVIREP	FISCSHLECR	TFFLTQGALL	NDKHSNGTVK	DRSPYRALMS	CPLGEAPSPY	NSKFESVAWS	ASACHDGMGW	LTIGISGPDN	

Figure 4 continued

	200									300
A/New Caledonia/20/99		e	r	v	g				t	
A/Thessaloniki/24/2005		e	r	v	g				t	
A/Netherlands/2/2007		e	r	v	g				t	
A/Massachusetts/5/2007		e	r	v	g			y	t	
A/Egypt/39/2005		e	r		g			s	t	
A/Solomon Islands/3/2006		e	r		g		t		t	
A/Fukushima/141/2006		e	r		g		t		t	
A/Hong Kong/2652/2006					g		m		t	
A/St. Petersburg/96/2007					r					
A/Brisbane/59/2007										
A/Barcelona/00083/2008										
A/Paris/0546/2007										
A/Berlin/98/2008										
A/Norway/1758/2007										
A/Belgium/G257/2008								y		
A/Paris/0341/2007								y		
A/Volos/108/2008								y		
A/Lisbon/3/2008								y		
A/Norway/1736/2007	a							y		
Consensus	GAVAVLKYN	IIITGTIKSWK	KQILRTQESE	CVCMNGSCFT	IMTDGPSNKA	ASYKIFKIEK	GKVTKSIELN	APNFHYEECS	CYPDTGIVMC	VCRDNWHGSN
	300									400
A/New Caledonia/20/99			k		d				d	
A/Thessaloniki/24/2005			k	k	d				d	
A/Netherlands/2/2007			k	k	d				d	
A/Massachusetts/5/2007			k	k	d				d	
A/Egypt/39/2005			k		n	d				
A/Solomon Islands/3/2006			k		d					
A/Fukushima/141/2006			k		d					
A/Hong Kong/2652/2006			k				i			i
A/St. Petersburg/96/2007			k		d	d				
A/Brisbane/59/2007						d				
A/Barcelona/00083/2008						d				
A/Paris/0546/2007						d				
A/Berlin/98/2008						d				
A/Norway/1758/2007						d				
A/Belgium/G257/2008										
A/Paris/0341/2007										
A/Volos/108/2008										
A/Lisbon/3/2008										
A/Norway/1736/2007										
Consensus	RPWVSFNQNL	DYQIGYICSG	VFGDNPRPED	GEGSCNPVT	DGANGVKGFS	YKYGNVWIG	RTKSNRLRKG	FEMIWDPNGW	TNTDSDFSVK	QDVVAITDWS

Figure 4 continued

	400						470
A/New Caledonia/20/99							
A/Thessaloniki/24/2005					d		
A/Netherlands/2/2007					d		
A/Massachusetts/5/2007					d		
A/Egypt/39/2005							
A/Solomon Islands/3/2006					g		
A/Fukushima/141/2006					g		
A/Hong Kong/2652/2006					i		
A/St. Petersburg/96/2007							
A/Brisbane/59/2007							
A/Barcelona/00083/2008							
A/Paris/0546/2007							
A/Berlin/98/2008							
A/Norway/1758/2007							
A/Belgium/G257/2008							
A/Paris/0341/2007							
A/Volos/108/2008							
A/Lisbon/3/2008							
A/Norway/1736/2007							
Consensus	GYSGSFVQHP	ELTGLDCIRP	CFWVELVRGL	PRENTTIWTS	GSSISFCGVN	SDTANWSWPD	GAELPFTIDK

Table 3. Amino acid changes characteristic of the recent principal H1N1 sequence variants

Clade	Representative strain	Amino acid changes	
		HA	NA
1 ¹	A/Thessaloniki/24/05	Y252F (T71I) (F260L) (D272G)	E332K N450D (N68S) (M188K) (D382N)
2 ¹	Common changes	T82K Y94H K140E R208K T266N	V234M D382N
2A ¹	A/Solomon Islands/3/06	K73R V128T R145K	M23I H64N G77E K173R S266T D452G
2B ¹	A/Brisbane/59/07	D35N R188K E273K	H55N K78E E214G R222Q G249K T287I K329E D344N (I38L) (H275Y) (G354D)
2C ¹	A/Hong Kong/2652/06	S36N R145K R188M A189T T193K (N244S)	S82P R130K M188I E214G R222Q I267M D344N L367I V393I T453I

1. Relative to A/New Caledonia/20/99, including common differences: HA, V165A and W251R; NA, V48I.

Table 4: Oseltamivir resistance of influenza A (H1N1) viruses.

Country	Viruses tested	Resistant to oseltamivir*	Proportion (%) resistant to oseltamivir
Austria	68	5	7
Belgium	4	1	25
Bulgaria	9	0	0
Croatia	6	0	0
Czech Republic	7	0	0
Denmark	46	2	4
Finland	11	3	27
France	279	106	38
Germany	395	46	12
Greece	44	4	9
Hungary	11	0	0
Ireland	46	5	11
Italy	92	1	1
Latvia	14	0	0
Luxembourg	154	39	25
Netherlands	89	22	27
Norway	184	127	69
Portugal	6	2	33
Romania	8	0	0
Slovakia	5	0	0
Slovenia	15	0	0
Spain	52	0	0
Sweden	16	2	12
Switzerland	44	6	14
United Kingdom	316	29	9
Total	1921	400	21

* IC₅₀ value >100 nM and/or H275Y mutation.

Figure 5. Amino acid substitutions characteristic of Clade 2B NAs of H1N1 viruses

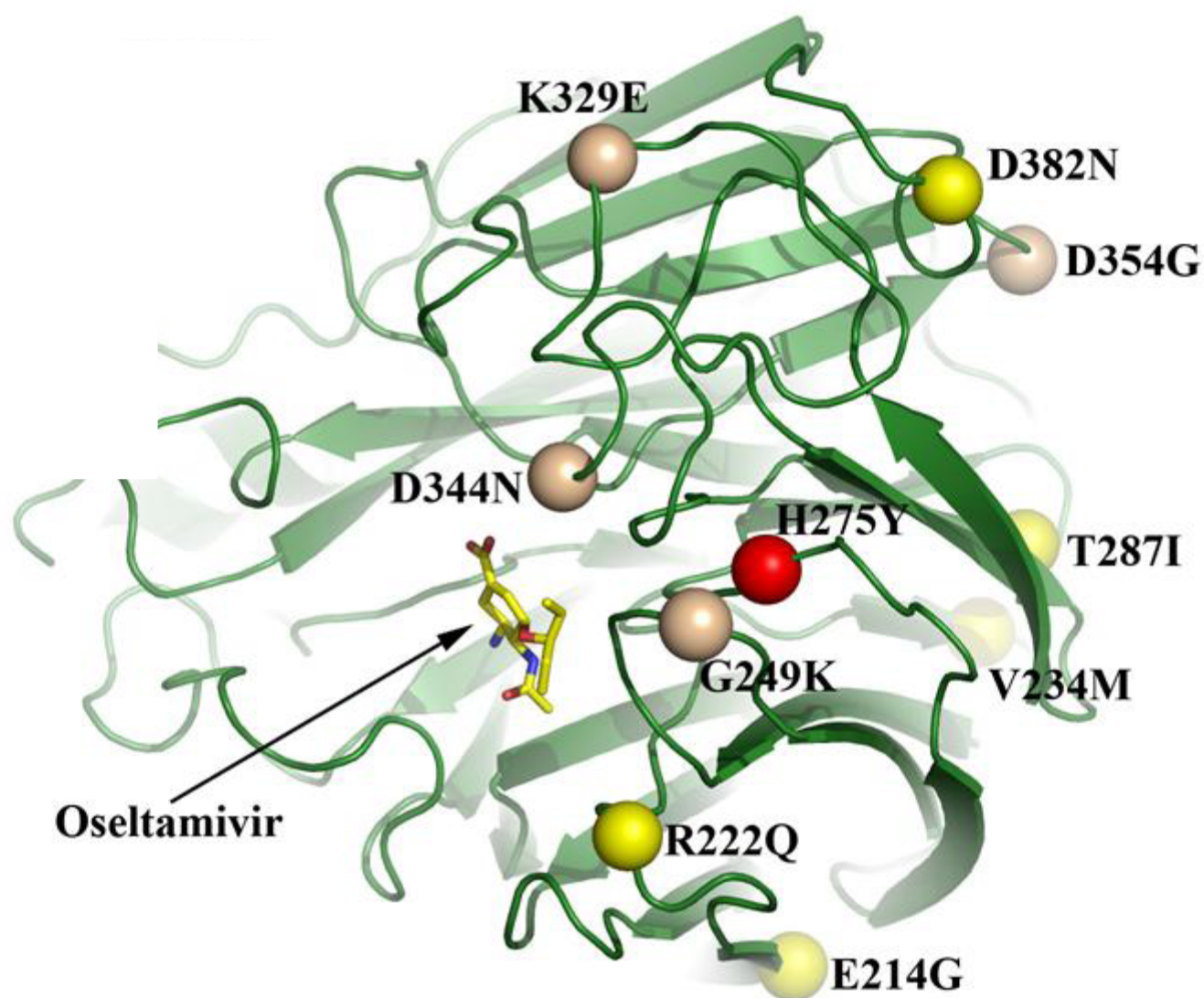


Table 5. Antigenic analyses of influenza A H3N2 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
			Post infection ferret sera										
			A/Wis	A/Nepal	A/Trieste	A/Sofia	A/Serr	A/Prag	A/Tri	A/Wis	A/Hen	A/Bris	A/Uru
			67/05	921/06	25c/07	319/07	77/07	3/07	25E/07	3/07	147/07	10/07	716/07
			F6/06	F2/07	F5/07	F6/07	F10/07	F9/07	F11/07	F15/07	F16/07	F17/07	F3/08
REFERENCE VIRUSES													
A/Wisconsin/67/2005	31/8/05	SpfCk3E3 \3	5120	2560	2560	1280	640	640	2560	640	2560	2560	2560
A/Nepal/921/2006	11/7/06	E2 \2	5120	5120	2560	2560	1280	1280	2560	640	5120	2560	2560
A/Trieste/25c/2007	Jan-07	MDCKx \2	160	80	80	80	80	<	160	160	80	320	160
A/Sofia/319/2007	Jan-07	E2 \2	5120	2560	2560	1280	640	640	1280	640	5120	1280	2560
A/Serres/77/2007	25/1/07	E1 \1	5120	5120	2560	2560	2560	1280	2560	1280	5120	2560	2560
A/Prague/3/2007	Feb-07	MDCK3, E4 \2	2560	5120	1280	1280	640	1280	1280	640	5120	2560	2560
A/Trieste/25E/2007	Jan-07	MDCKx \2,E1	5120	2560	1280	1280	320	640	1280	640	5120	1280	2560
A/Wisconsin/3/2007	21/1/07	Ex \2	5120	5120	5120	2560	2560	1280	2560	2560	5120	5120	5120
A/Henan/Jinshui/147/2007	unknown	E3 \1	1280	2560	1280	1280	320	640	1280	640	5120	1280	2560
A/Brisbane/10/2007	6/2/07	E2 \3	1280	2560	1280	1280	320	640	1280	640	5120	1280	2560
A/Uruguay/716/2007	21/6/07	spfck1, E3 \2	2560	2560	1280	1280	640	640	1280	640	5120	1280	5120
TEST VIRUSES													
A/Kumamoto/30/2007	unknown	MDCK1+2 \1	160	40	80	80	80	<	80	160	80	320	160
A/Toyama/119/2007	unknown	MDCK1+2 \1	160	80	80	40	40	<	80	80	80	160	160
A/Madagascar/1236/2007	06/08/07	MDCK2 \2	2560	5120	2560	1280	320	640	2560	640	5120	2560	—
A/Texas/37/2007	13/10/07	MK1,MDCK3 \1	2560	640	640	640	320	80	640	640	640	1280	640
A/Nevada/05/2007	05/10/07	x,MDCK2 \1	80	80	80	80	80	<	160	160	80	320	160
A/Taiwan/760/2007	04/10/07	MDCK5 \1	320	160	160	160	160	<	320	320	160	640	160
A/Paris/0259/2007	Nov-08	MDCKx \1	80	40	40	40	40	<	160	80	40	320	160
A/Paris/0358/2007	Nov-08	MDCKx \1	160	80	80	80	80	<	160	160	80	320	160
A/Switzerland/6904/2007	14/11/07	Cx \1	160	80	80	80	80	<	160	160	80	320	—
A/Paris/0653/2007	Dec-07	MDCKx \1	160	80	80	80	80	<	160	160	80	320	—
A/Belgium/SP13/2008	04/12/07	MDCKx \1	80	80	80	40	40	<	80	80	80	320	160
A/Belgium/SP12/2008	06/12/07	MDCKx \1	160	80	80	80	80	<	80	80	160	320	160
A/Hong Kong/4761/2007	17/12/07	MDCK2 \1	80	80	80	40	40	<	80	160	80	160	—
A/Lyon/1381/2007	18/12/07	MDCK2 \1	640	80	80	160	160	<	160	160	160	320	160
A/Hong Kong/4765/2007	22/12/07	MDCK2 \1	160	160	80	160	160	80	80	160	80	160	—
A/Stockholm/2/2008	Jan-08	MDCK1 \1	80	40	<	40	40	<	80	80	<	80	80
A/Finland/9/2008	07/01/08	MDCK2 \1	160	40	80	80	<	<	80	80	80	160	320
A/Novosibirsk/1/2008	13/02/08	C1 \1	160	80	160	80	80	<	160	160	80	320	160

1. <, <40

Vaccine strain

Figure 6. Phylogenetic comparison of H3 HA genes

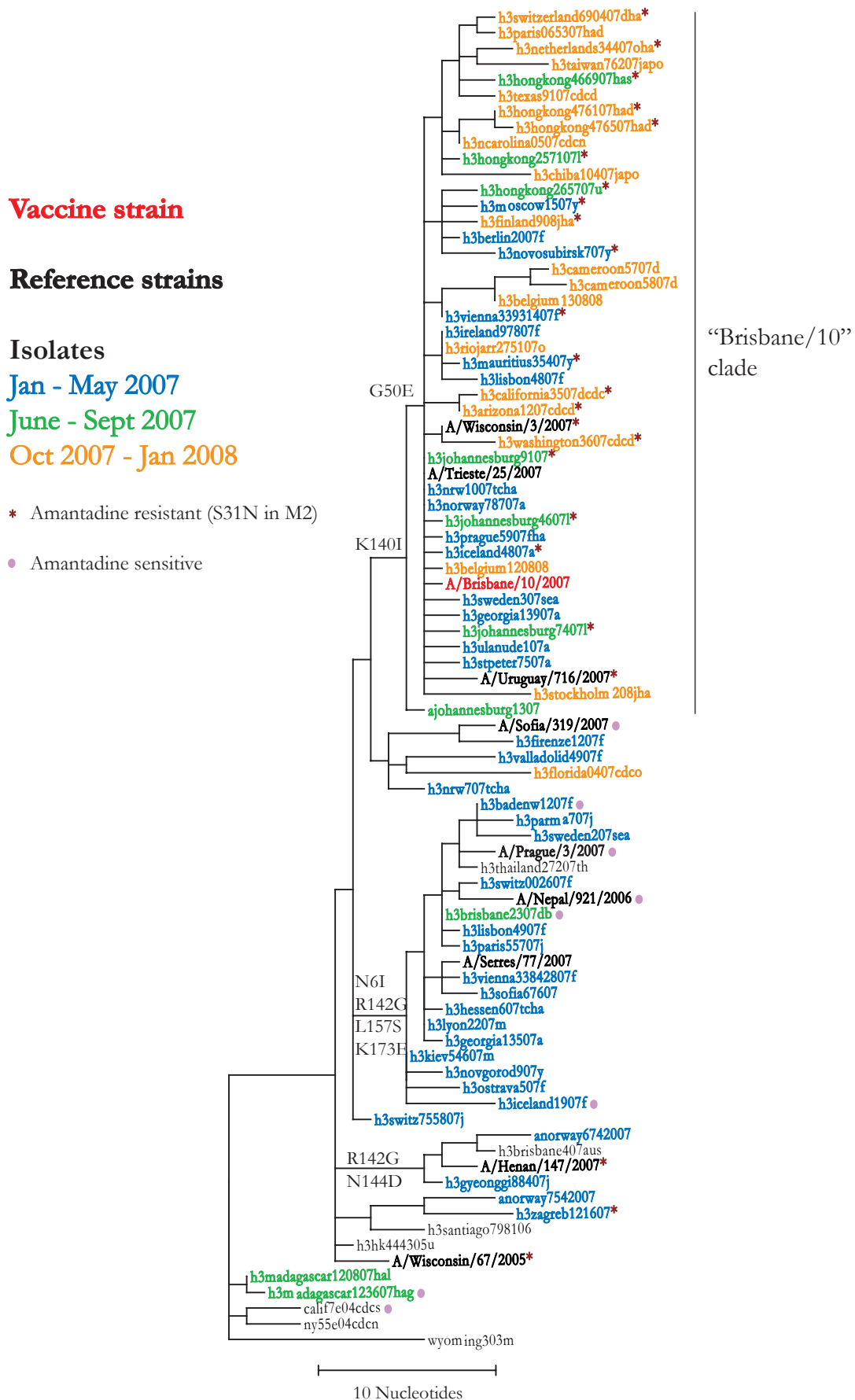


Figure 7. HA1 sequences of H3N2 viruses

	1									100	
A/California/7/2004						g					
A/Wisconsin/67/2005						g					
A/Zagreb/1217/2007											
A/Sofia/319/2007						g					
A/Henan/147/2007	y					g					
A/Prague/3/2007	i				n	g			l		
A/Serres/77/2007	r	i	v			g					
A/Nepal/921/2006	i										
A/Hong Kong/4669/2007							t				
A/Paris/0653/2007											
A/Switzerland/6904/2007											
A/Netherlands/344/2007											
A/Brisbane/10/2007											
A/Trieste/25/2007											
A/Stockholm/2/2008	f							i			
A/Finland/9/2008											
A/Wisconsin/3/2007											
A/Uruguay/716/2007											
A/Belgium/1308/2008											
A/Cameroon/58/2007						n			e		
Consensus	QKLPGNDNST	ATLCLGHHAV	PNGTIVKTIT	NDQIEVTNAT	ELVQSSSTGE	ICDSPHQILD	GENCTLIDAL	LGDPQCDGFQ	NKKWDLFVER	SKAYSNCYPY	
	101									200	
A/California/7/2004				s	k				n	s	t
A/Wisconsin/67/2005			d		k						h
A/Zagreb/1217/2007			a		k			d		p	t
A/Sofia/319/2007					k					p	
A/Henan/147/2007					k	g	d			p	
A/Prague/3/2007					k	g		s	e		p
A/Serres/77/2007			a		k	g		s	e	v	
A/Nepal/921/2006			a		k	g		s	e	l	
A/Hong Kong/4669/2007								s	e		
A/Paris/0653/2007								s			
A/Switzerland/6904/2007			a					s			
A/Netherlands/344/2007								s			
A/Brisbane/10/2007									a	s	
A/Trieste/25/2007											
A/Stockholm/2/2008			d						n		
A/Finland/9/2008											
A/Wisconsin/3/2007									e		
A/Uruguay/716/2007					s						p
A/Belgium/1308/2008								r			
A/Cameroon/58/2007								r			
Consensus	DVPDYASLRS	LVASSGTLEF	NNESFNWTGV	TQNGTSSACI	RRSNNSFFSR	LNWLTHLKFK	YPALNVTMPN	NEKFDKLYIW	GVHHPGTDND	QIFLYAQASG	

Figure 7 continued

	201									300
A/California/7/2004			d							
A/Wisconsin/67/2005			i							
A/Zagreb/1217/2007										
A/Sofia/319/2007								s		
A/Henan/147/2007										
A/Prague/3/2007										
A/Serres/77/2007										
A/Nepal/921/2006										
A/Hong Kong/4669/2007										
A/Paris/0653/2007										
A/Switzerland/6904/2007										
A/Netherlands/344/2007										n
A/Brisbane/10/2007										
A/Trieste/25/2007										
A/Stockholm/2/2008			i		t					
A/Finland/9/2008							q			
A/Wisconsin/3/2007										
A/Uruguay/716/2007										
A/Belgium/1308/2008										
A/Cameroon/58/2007										
Consensus	RITVSTKRSQ	QTVIPNIGSR	PRVRNIPSRI	SIYWTIVKPG	DILLINSTGN	LIAPRGYFKI	RSGKSSIMRS	DAPIGKCNSE	CITPNGSIPN	DKPFQNVNRI
	301		339							
A/California/7/2004										
A/Wisconsin/67/2005										
A/Zagreb/1217/2007										
A/Sofia/319/2007										
A/Henan/147/2007										
A/Prague/3/2007										
A/Serres/77/2007		t								
A/Nepal/921/2006										
A/Hong Kong/4669/2007										
A/Paris/0653/2007										
A/Switzerland/6904/2007										
A/Netherlands/344/2007										
A/Brisbane/10/2007										
A/Trieste/25/2007										
A/Stockholm/2/2008										
A/Finland/9/2008										
A/Wisconsin/3/2007										
A/Uruguay/716/2007										
A/Belgium/1308/2008										
A/Cameroon/58/2007										
Consensus	TYGACPRYVK	QNTLKLATGM	RNVPEKQTR							

Figure 8. Phylogenetic comparison of N2 neuraminidase genes

Vaccine strain

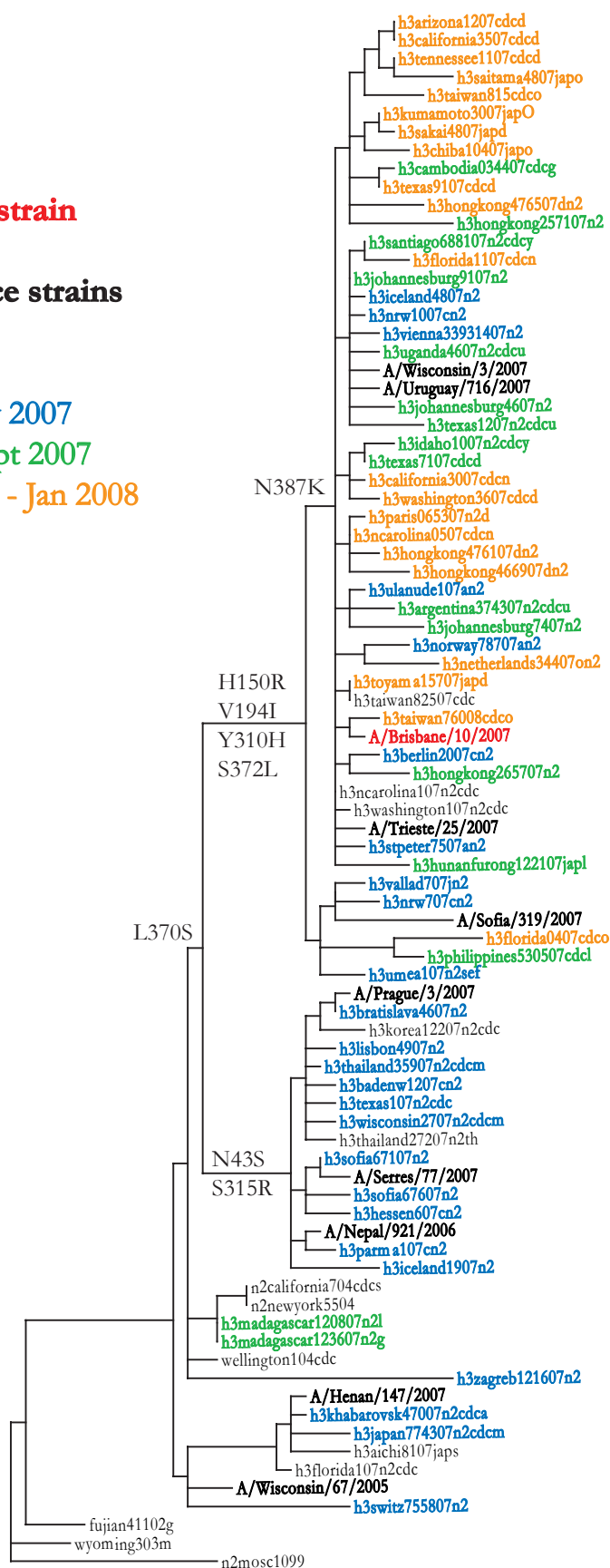
Reference strains

Isolates

Jan - May 2007

June - Sept 2007

Oct 2007 - Jan 2008



"Brisbane 10"
clade

10 Nucleotides

Figure 9. Neuraminidase sequences of H3N2 viruses

	1									100
A/Wyoming/3/2003										n
A/California/7/2004										
A/Wisconsin/67/2005										n
A/Henan/147/2007								k		n
A/Zagreb/1216/2007										
A/Nepal/921/2006				s						
A/Prague/3/2007				p						
A/Serres/77/2007				s						
A/Sofia/319/2007						t				
A/Brisbane/10/2007										
A/Johannesburg/74/2007			v		m					n
A/Trieste/25/2007										
A/Netherlands/344/2007	v									
A/Hong Kong/4669/2007										
A/Hong Kong/4765/2007										
A/Paris/0653/2007										
A/Wisconsin/3/2007			t							
A/Uruguay/716/2007			t							
A/Vienna/339314/2007			t							
Consensus	MNPNQKIITI	GSVSLTISTI	CFFMQIAILI	TTVTLHFKQY	EFNSPPNNQV	MLCEPTIER	NITEIVYLTN	TTIEKEICPK	LAEYRNWSKP	QCDITGFAPP
	100									200
A/Wyoming/3/2003					h					v
A/California/7/2004					h					e
A/Wisconsin/67/2005					h					v
A/Henan/147/2007										v
A/Zagreb/1216/2007					h					v
A/Nepal/921/2006					h					v
A/Prague/3/2007					h					v
A/Serres/77/2007					h					v
A/Sofia/319/2007										
A/Brisbane/10/2007										
A/Johannesburg/74/2007										
A/Trieste/25/2007						g				
A/Netherlands/344/2007										
A/Hong Kong/4669/2007										
A/Hong Kong/4765/2007					n					
A/Paris/0653/2007										
A/Wisconsin/3/2007						e				
A/Uruguay/716/2007					h					
A/Vienna/339314/2007						n				
Consensus	SKDNSIRLSA	GGDIWVTREP	YVSCDPDKCY	QFALGQGTTL	NNVHSNDTVR	DRTPYRTLML	NELGVPFHLG	TKQVCIWSS	SSCHDGKAWL	HVCITGDDKN

Figure 9 continued

	200									300
A/Wyoming/3/2003		g	k							
A/California/7/2004										
A/Wisconsin/67/2005									l	
A/Henan/147/2007										r
A/Zagreb/1216/2007		v	d		n					
A/Nepal/921/2006										
A/Prague/3/2007										
A/Serres/77/2007										
A/Sofia/319/2007							ff			
A/Brisbane/10/2007										
A/Johannesburg/74/2007										
A/Trieste/25/2007										
A/Netherlands/344/2007										
A/Hong Kong/4669/2007		v								
A/Hong Kong/4765/2007										
A/Paris/0653/2007		v								
A/Wisconsin/3/2007										
A/Uruguay/716/2007							i			
A/Vienna/339314/2007										
Consensus	ATASFIYNGR	LVDSIVWSK	EILRTQESEC	VCINGTCTVV	MTDGSASGKA	DTKILFIEEG	KIVHTSTLSG	SAQHVEECSC	YPRYPGVRCV	CRDNWKGSNR
	300									400
A/Wyoming/3/2003	y						l	s	n	
A/California/7/2004	y						l	s	n	
A/Wisconsin/67/2005	y						l	s	n	
A/Henan/147/2007	m y			g			l	s	n	
A/Zagreb/1216/2007	ny		t				l		n	k
A/Nepal/921/2006	y	r						s	n	
A/Prague/3/2007	y	r						s	n	
A/Serres/77/2007	y	r						s	n	
A/Sofia/319/2007									n	
A/Brisbane/10/2007		t								
A/Johannesburg/74/2007										
A/Trieste/25/2007										
A/Netherlands/344/2007						l	l	f		
A/Hong Kong/4669/2007								i		
A/Hong Kong/4765/2007	r	r								
A/Paris/0653/2007										
A/Wisconsin/3/2007										
A/Uruguay/716/2007										
A/Vienna/339314/2007										
Consensus	PIVDINIKDH	SIVSSYVCSG	LVGDTPRKND	SSSSSHCLDP	NNEEGGHGVK	GWAFDDGNDV	WMGRTISEKS	RLGYETFKVI	EGWSNPKSKL	QINRQVIVDR

Figure 9 continued

	400						469
A/Wyoming/3/2003				_q			
A/California/7/2004							
A/Wisconsin/67/2005							
A/Henan/147/2007							
A/Zagreb/1216/2007							
A/Nepal/921/2006							
A/Prague/3/2007							
A/Serres/77/2007							
A/Sofia/319/2007							
A/Brisbane/10/2007							
A/Johannesburg/74/2007							
A/Trieste/25/2007							
A/Hong Kong/4669/2007							
A/Hong Kong/4765/2007							
A/Paris/0653/2007							
A/Wisconsin/3/2007							
A/Uruguay/716/2007							
A/Vienna/339314/2007	S						
Consensus	GNRSGYSGIF	SVEGKSCINR	CFYVELIRGR	KEETEVLTWS	NSIVVFCGTS	GTYGTGSWPD	GADINLMPI

Table 6. Amino acid changes characteristic of the principal H3N2 sequence variants

Clade	Representative strain	Amino Acid changes ¹	
		HA	NA
A	A/Brisbane/10/07	G50E K140I (T131N) (L157S) (K158R) (K173Q) (R261Q)	H150R V194I Y310H L370S S372L N387K
B	A/Nepal/921/06	N6I R142G L157S K173E (T128A)(-CHO)	N43S S315R L370S

1. Relative to A/Wisconsin/67/05. ± CHO indicates addition or loss of a glycosylation site.

Table 7. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Sich	B/Jiang	B/Eg	B/FI	B/FI	B/Bris	B/Sen
			379/99 F19/01	10/03 F8/04	144/05 F3/07	7/04 F21/05	4/06 F20/07	3/07 F24/07	114/07 F1/08
REFERENCE VIRUSES									
B/Sichuan/379/99	unknown	Ex	160	20	320	160	320	640	320
B/Jiangsu/10/2003	12.9.03	E4 \3	<	160	40	<	<	<	<
B/Egypt/144/2005	5.1.05	E2 \2	80	20	320	160	320	640	320
B/Florida/7/2004	1.10.05	E3 \2	160	40	320	320	640	640	640
B/Florida/4/2006	15.12.06	E2 \1	160	40	320	320	640	640	320
B/Brisbane/3/2007	9.3.07	E2 \2	80	20	320	320	320	640	320
B/Sendai-H/114/2007	unknown	E1+2 \1	80	20	160	160	320	320	320
TEST VIRUSES									
B/Stockholm/4/2007	07/11/07	MDCK0 \1	<	40	160	40	40	40	80
B/England/555/2007	Dec-07	Cx \1	<	10	160	<	40	<	80
B/Ireland/v12143/2007	Dec-07	Cx \1	80	160	640	320	160	160	320
B/Paris/0584/2007	Dec-07	MDCKx \1	40	40	160	80	640	80	—
B/Finland/1/2007	01/12/07	MDCK2 \1	<	20	80	40	80	40	160
B/Lyon/1283/2007	07/12/07	MDCK2 \1	40	80	320	160	160	160	—
B/Thuringen/2/2007	10/12/17	MDCKx \1	40	80	320	80	160	640	—
B/Austria/399638/2007	19/12/07	MDCK2 \1	40	40	160	80	320	160	—
B/Hong Kong/1360/2007	22/12/07	MDCK2 \1	80	80	320	80	320	160	—
B/Belgium/G212/2008	24/12/07	MDCKx \1	160	320	640	640	320	320	320
B/Zagreb/2513/2008	Jan-08	Ex \1	40	10	80	80	80	160	160
B/Norway/51/2008	01/01/08	MDCK1 \1	80	80	320	320	320	320	—
B/Berlin/1/2008	03/01/08	MDCKx \1	80	80	320	320	320	320	—
B/Switzerland/7946/2008	03/01/08	Cx \1	<	<	160	80	80	80	—
B/Barcelona/00202/2008	06/01/08	MDCK0 \1	<	40	160	80	80	80	320
B/Denmark/3/2008	07/01/08	MDCK2 \1	<	40	160	80	80	80	320
B/Sachsen/2/2008	11/01/08	MDCKx \1	40	80	160	160	160	160	160
B/Praha/77/2008	17/01/08	MDCK1 \1	<	160	320	160	160	160	160
B/Perugia/9/2008	24/01/08	MDCK1 \1	<	40	320	80	160	160	160
B/Lasi/474/2008	30/01/08	MDCK1,E1 \2	<	10	40	80	40	80	160
B/Roma-ISS/3/2008	31/01/08	MDCK2 \1	80	320	320	320	320	160	320
B/Parma/30/2008	04/02/08	MDCKx \1	40	80	160	160	160	320	320
B/Lisbon/12/2008	05/02/08	MDCK2 \1	80	160	320	320	160	320	320
B/Valladolid/15/2008	06/02/08	MDCK1 \1	160	320	640	640	320	640	640
B/Parma/35/2008	06/02/08	MDCKx \1	<	80	320	80	160	80	80
B/Salamanca/17/2008	11/02/08	MDCK1 \1	40	80	160	160	160	160	320
B/Ekaterinburg/1/2008	15/02/08	C2 \1	40	80	160	160	80	160	160

¹ <, <40

Vaccine strain

Figure 10. Phylogenetic comparison of B HA genes (Yamagata lineage)

Vaccine strains

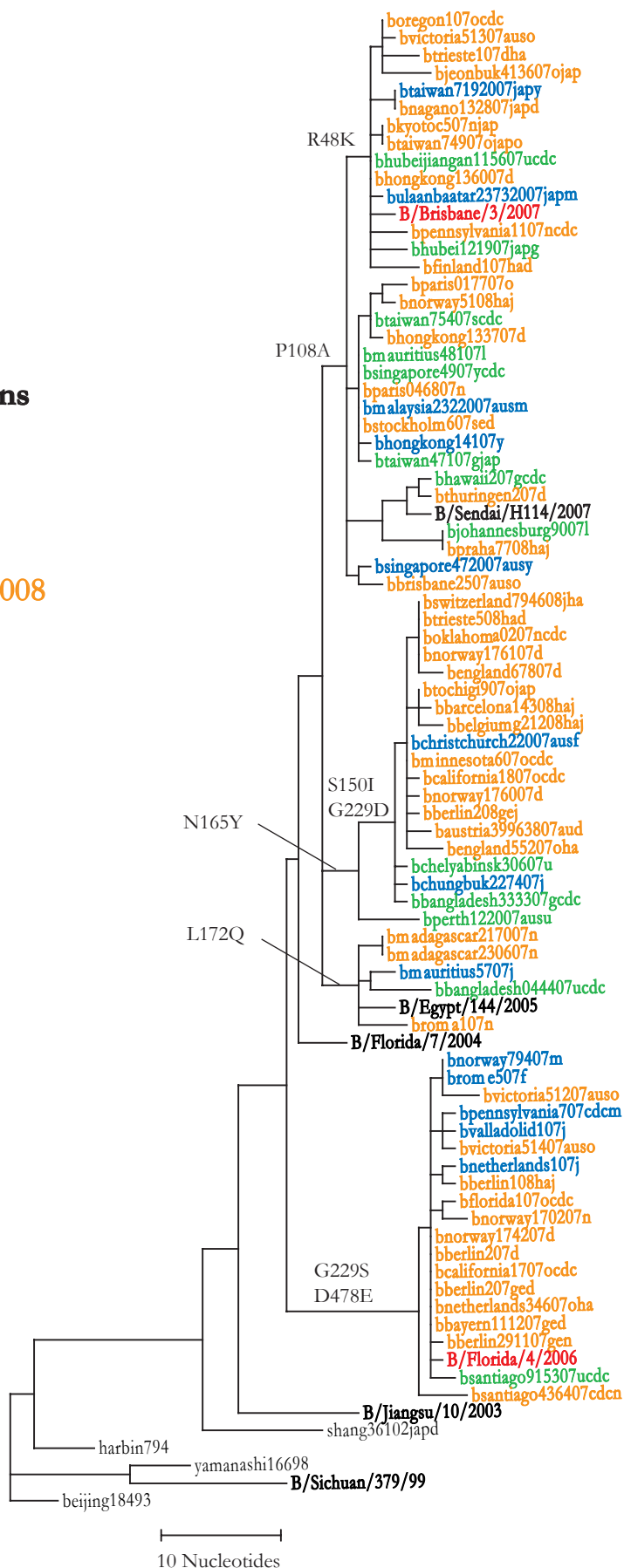
Reference strains

Isolates

Jan - May 2007

June - Sept 2007

Oct 2007 - Jan 2008



	1									100
B/Sichuan/379/99			a	h	k	t		i	ik	
B/Shanghai/361/2002				i	h					
B/Jiangsu/10/2003										
B/Florida/7/2004										
B/Egypt/144/2005									i	
B/Norway/51/2008										
B/Sendai/H114/2007										
B/Brisbane/3/2007					k					
B/Praha/77/2008								i		
B/Finland/1/2007					k		m			
B/Thuringen/2/2007										
B/Paris/0468/2007										
B/Hong Kong/1337/2007										
B/Trieste/5/2008										
B/Barcelona/143/2008										
B/England/552/2007							l			
B/Austria/399638/2007										
B/Florida/4/2006									k	
B/Netherlands/346/2007										
B/Berlin/1/2008										
Consensus	DRICTGITSS	NSPHVVKTAT	QGEVNVTVGI	PLTTTPTKSY	FANLKGTRTR	GKLCPCDCLNC	TOLDVALGRP	MCVGTTPSAK	ASILHEVRPV	TSGCFPIMHD
	100									200
B/Sichuan/379/99		k	n				r	t	h	ke
B/Shanghai/361/2002				l						d
B/Jiangsu/10/2003			n		r				v	
B/Florida/7/2004										d
B/Egypt/144/2005								q		d
B/Norway/51/2008	a									n
B/Sendai/H114/2007	a									x
B/Brisbane/3/2007	a								q	d
B/Praha/77/2008	a									
B/Finland/1/2007	a									
B/Thuringen/2/2007	a									
B/Paris/0468/2007	a									
B/Hong Kong/1337/2007	a									
B/Trieste/5/2008						i	y			
B/Barcelona/143/2008						i	y			
B/England/552/2007						i	y			
B/Austria/399638/2007						i	y			
B/Florida/4/2006										
B/Netherlands/346/2007										
B/Berlin/1/2008										
Consensus	RTKIRQLPNL	LRGYENIRLS	TONVIDAEKA	PGGPYRLGTS	GSCPNATSKS	GFFATMAWAV	PKDNNKNATN	PLTVEVPYIC	TEGEDQITVM	GFHSDNKTQM

Figure 11 continued

	200									300
B/Sichuan/379/99		i				v		i		
B/Shanghai/361/2002						v				
B/Jiangsu/10/2003			a	e		v	r			
B/Florida/7/2004										
B/Egypt/144/2005										
B/Norway/51/2008										
B/Sendai/H114/2007										
B/Brisbane/3/2007										
B/Praha/77/2008										
B/Finland/1/2007										
B/Thuringen/2/2007										
B/Paris/0468/2007										
B/Hong Kong/1337/2007										
B/Trieste/5/2008			d							
B/Barcelona/143/2008			d							
B/England/552/2007			d							
B/Austria/399638/2007			d							
B/Florida/4/2006			s							
B/Netherlands/346/2007			s							
B/Berlin/1/2008			s							
Consensus	KNLYGDSNPQ	KFTSSANGVT	THYVSQIGGF	PDQTEDGGLP	QSGRIVVDYM	MQKPGKTGTI	VYQRGVLLPQ	KWVCASGRSK	VIKGSPLIG	EADCLHEKYG
	300									
B/Sichuan/379/99										
B/Shanghai/361/2002										
B/Jiangsu/10/2003										
B/Florida/7/2004										
B/Egypt/144/2005										
B/Norway/51/2008										
B/Sendai/H114/2007										
B/Brisbane/3/2007										
B/Praha/77/2008										
B/Finland/1/2007										
B/Thuringen/2/2007										
B/Paris/0468/2007										
B/Hong Kong/1337/2007										
B/Trieste/5/2008										
B/Barcelona/143/2008										
B/England/552/2007										
B/Austria/399638/2007										
B/Florida/4/2006										
B/Netherlands/346/2007										
B/Berlin/1/2008										
Consensus	GLNKSPPYYT	GEHAKAIGNC	PIWVKTPLKL	ANGTKYRPPA	KLLKER					

Figure 12. Phylogenetic comparison of B neuraminidase genes

Vaccine strains

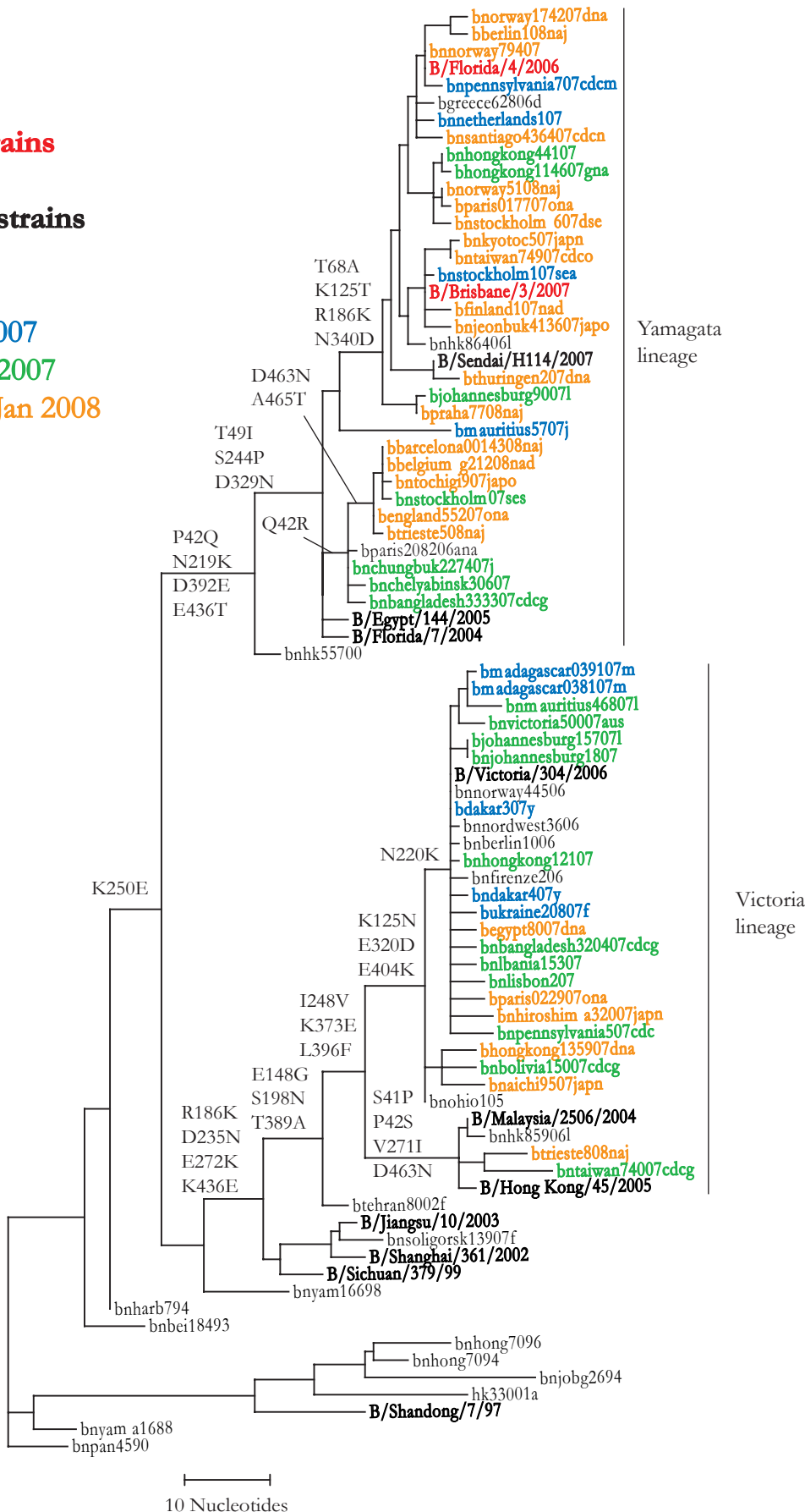
Reference strains

Isolates

Jan - May 2007

June - Sept 2007

Oct 2007 - Jan 2008



	1									100						
B/Shandong/7/97				l	p	kvi	t	s		h	ee	m	f		q	
B/Sichuan/379/99					p		t						f			
B/Shanghai/361/2002						t		t								
B/Jiangsu/10/2003						t	t	t								
B/Malaysia/2506/2004						ps		t								
B/Hong Kong/45/2005						ps		t								
B/Trieste/8/2008						ps		t								
B/Victoria/304/2006						p		t								
B/Hong Kong/1359/2007						p		t		h						
B/Johannesburg/157/2007						p		t								
B/Paris/0229/2007						p		t								
B/Dakar/4/2007						p	n	t								
B/Florida/7/2004								i		h						
B/Egypt/144/2005					r			i								
B/England/552/2007						r		i								
B/Trieste/5/2008						r		i								
B/Barcelona/00143/2008						r		i								
B/Florida/4/2006								i			a					
B/Brisbane/3/2007	k							i			a					
B/Sendai/H114/2007				x				i			a					
B/Berlin/1/2008								i			a					
B/Praha/77/2008								i			a					
B/Norway/51/2008								i			a					
B/Finland/1/2007								i			a					
Consensus	MLPSTIQTLT	LFLTSGGVLL	SLYVSASLSY	LLYSDILLKF	SQTEITAP-M	PLDCANASNV	QAVNRSATKG	VTLLLPEPEW	TYPRLSCP GS	TFQKALLISP						
	100															200
B/Shandong/7/97	a							n							r	
B/Sichuan/379/99																
B/Shanghai/361/2002																
B/Jiangsu/10/2003																
B/Malaysia/2506/2004								g								n
B/Hong Kong/45/2005								g								n
B/Trieste/8/2008								g								n
B/Victoria/304/2006				n				g								n
B/Hong Kong/1359/2007				n				g								n
B/Johannesburg/157/2007				n				g								n
B/Paris/0229/2007				n				g								n
B/Dakar/4/2007				n				g								n
B/Florida/7/2004															r	
B/Egypt/144/2005															r	
B/England/552/2007															r	
B/Trieste/5/2008															r	n
B/Barcelona/00143/2008															r	
B/Florida/4/2006				t												
B/Brisbane/3/2007				t												
B/Sendai/H114/2007				t												v
B/Berlin/1/2008																

Figure 13 continued

	200									300
B/Shandong/7/97		n		s	r			e		
B/Sichuan/379/99		n	n	s						
B/Shanghai/361/2002		n	n	s						
B/Jiangsu/10/2003		n	n	s						
B/Malaysia/2506/2004		n	n	s	v			i		
B/Hong Kong/45/2005		n	n	s	v			i		
B/Trieste/8/2008		n	n	s	v			i		
B/Victoria/304/2006		nk	n	s	v					
B/Hong Kong/1359/2007		n	n	s	v		v			
B/Johannesburg/157/2007		nk	n	s	v					
B/Paris/0229/2007		nk	n	s	v		t			
B/Dakar/4/2007		nk	n	s	v				e	
B/Florida/7/2004		k		p						
B/Egypt/144/2005		k		p						
B/England/552/2007		k		p						
B/Trieste/5/2008		k		p						
B/Barcelona/00143/2008		k		p						
B/Florida/4/2006		k		p						
B/Brisbane/3/2007		k		p	v					
B/Sendai/H114/2007		k		p						
B/Berlin/1/2008		k		p						r
B/Praha/77/2008		k		p					n	
B/Norway/51/2008		k		p						
B/Finland/1/2007		k		p	v					
Consensus	LLKIKYGEAY	TDYHSYA-N	ILRTQESACN	CIGGDCYLM	TDG-ASGISE	CRFLKIREGR	IIKEIFPTGR	VKHTTEECTG	FASNKTIECA	CRDNSYTAKR
	300									400
B/Shandong/7/97			d		n	r		e		p
B/Sichuan/379/99			d							
B/Shanghai/361/2002			d		n				i	
B/Jiangsu/10/2003			d		n					
B/Malaysia/2506/2004			d						a	f
B/Hong Kong/45/2005			d					e	a	tf
B/Trieste/8/2008			d		g			e_k	a	vf
B/Victoria/304/2006		d	d					e	a	f
B/Hong Kong/1359/2007		d	d					e	a	f
B/Johannesburg/157/2007		d	d					e	a	f
B/Paris/0229/2007		d	d					e	a	f
B/Dakar/4/2007		d	d					e	a	f
B/Florida/7/2004			n							e
B/Egypt/144/2005			n					r		e
B/England/552/2007			n							e
B/Trieste/5/2008			n							e
B/Barcelona/00143/2008			n							e
B/Florida/4/2006			n	d						e
B/Brisbane/3/2007			n	d						e
B/Sendai/H114/2007			n	g_d						e
B/Berlin/1/2008			n	d						g
B/Praha/77/2008			n	d						e
B/Norway/51/2008			n	d						e
B/Finland/1/2007			n	d						e
Consensus	PFVKLVNVED	TAEIRLMCTE	TYLDTPRP-D	GSITGPCESN	GDKGSGGIKG	GFVHQRMASK	IGRWYSRTMS	KTKRMGMGLY	VKYDGDPTWD	SDALALSGVM

Figure 13 continued

	400						486
B/Shandong/7/97	i			k			
B/Sichuan/379/99				e			
B/Shanghai/361/2002				e			
B/Jiangsu/10/2003				e			
B/Malaysia/2506/2004				e			n
B/Hong Kong/45/2005				e			n
B/Trieste/8/2008				e			n
B/Victoria/304/2006	k			e			
B/Hong Kong/1359/2007	k			e			
B/Johannesburg/157/2007	k			e			
B/Paris/0229/2007	k			e			
B/Dakar/4/2007	k			e			
B/Florida/7/2004							
B/Egypt/144/2005							
B/England/552/2007							n_t
B/Trieste/5/2008							n_t
B/Barcelona/00143/2008							n_t
B/Florida/4/2006							
B/Brisbane/3/2007							
B/Sendai/H114/2007	d						
B/Berlin/1/2008	i						
B/Praha/77/2008							
B/Norway/51/2008							
B/Finland/1/2007							
Consensus	VSMEEPGWYS	FGFEIKDKKC	DVPCIGIEMV	HDGGKTTWHS	AATAIYCLMG	SGQLLWDTVT	GVDMAL

Table 8. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Yamagata ¹	B/England/552/07	N165Y S150I G229D	Q42R D463N A465T
	B/Brisbane/3/07	P108A (R48K)	T68A K125T R186K N340D
	B/Florida/4/06	G229S D478E	T68A K125T R186K N340D
Victoria ²	B/Victoria/304/06	(K56Q) (N165K)	P41S S42P K125N N220K I271V E320D E404K N463D

1. Relative to B/Florida/7/04. 2. Relative to B/Malaysia/2506/04.

Table 9. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Isolation date	Passage History	Haemagglutination inhibition titre ¹					
			B/Shan ² 7/97 SH 392	Post infection ferret sera				
				B/Shan 7/97 F12/02	B/Bris ³ 32/02 F2/03	B/HK 45/05 F15/05	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06
REFERENCE VRUSES								
B/Shandong/7/97		Ex	2560	160	80	80	160	160
B/Brisbane/32/2002	02/07/02	E2 \6	1280	160	80	160	160	80
B/HK/45/2005	07/02/05	MDCK2 \4	2560	320	160	160	320	160
B/Malaysia/2506/2004	01/12/04	E3 \3	2560	320	320	160	320	320
B/Victoria/304/2006	06/06/06	E2 \2	1280	160	160	160	320	320
TEST VIRUSES								
B/Paris/0229/2007	Oct-07	MDCKx \1	640	<	<	<	40	<
B/Hiroshima/320/2007	Nov-07	E1+2 \1	1280	160	160	160	160	160
B/Switzerland/7005/2007	19/11/07	MDCK2 \1	2560	40	40	320	40	40
B/Egypt/80/2007	05/12/07	E1 am \1	1280	160	80	80	160	160
B/Egypt/80/2007	05/12/07	E2 all \1	2560	160	80	80	320	320
B/Hong Kong/1328/2007	07/12/07	MDCK2 \1	1280	<	<	160	<	<
B/Hong Kong/1332/2007	10/12/07	MDCK2 \1	640	<	<	160	<	<
B/Trieste/03/2007	12/12/07	MDCKsiat1 \1	2560	40	<	160	<	<
B/Trieste/04/2007	17/12/07	MDCKsiat1 \1	2560	80	<	160	40	<
B/Trieste/06/2007	19/12/07	MDCKsiat1 \1	2560	<	<	320	<	<
B/Hong Kong/1359/2007	22/12/07	MDCK2 \1	1280	40	40	160	160	80
B/Denmark/1/2008	04/01/08	MDCK2 \1	5120	40	40	320	80	160
B/Trieste/6/2008	14/01/08	MDCK1 \1	1280	<	<	160	<	<
B/Trieste/8/2008	21/01/08	MDCK1 \1	1280	<	<	160	<	<
B/Valladolid/11/2008	23/01/08	MDCK1 \1	1280	80	<	320	40	40
B/Parma/25/2008	29/01/08	MDCK \1	1280	80	<	320	40	40
B/Parma/36/2008	6/2/2008	MDCK \1	1280	160	<	320	40	40

1. <, <40; 2. hyperimmune sheep serum, 3. <, <10

Figure 14. Phylogenetic comparison of B HA genes (Victoria lineage)

Vaccine strain

Reference strains

Isolates

Jan - May 2007

June - Sept 2007

Oct 2007 - Jan 2008

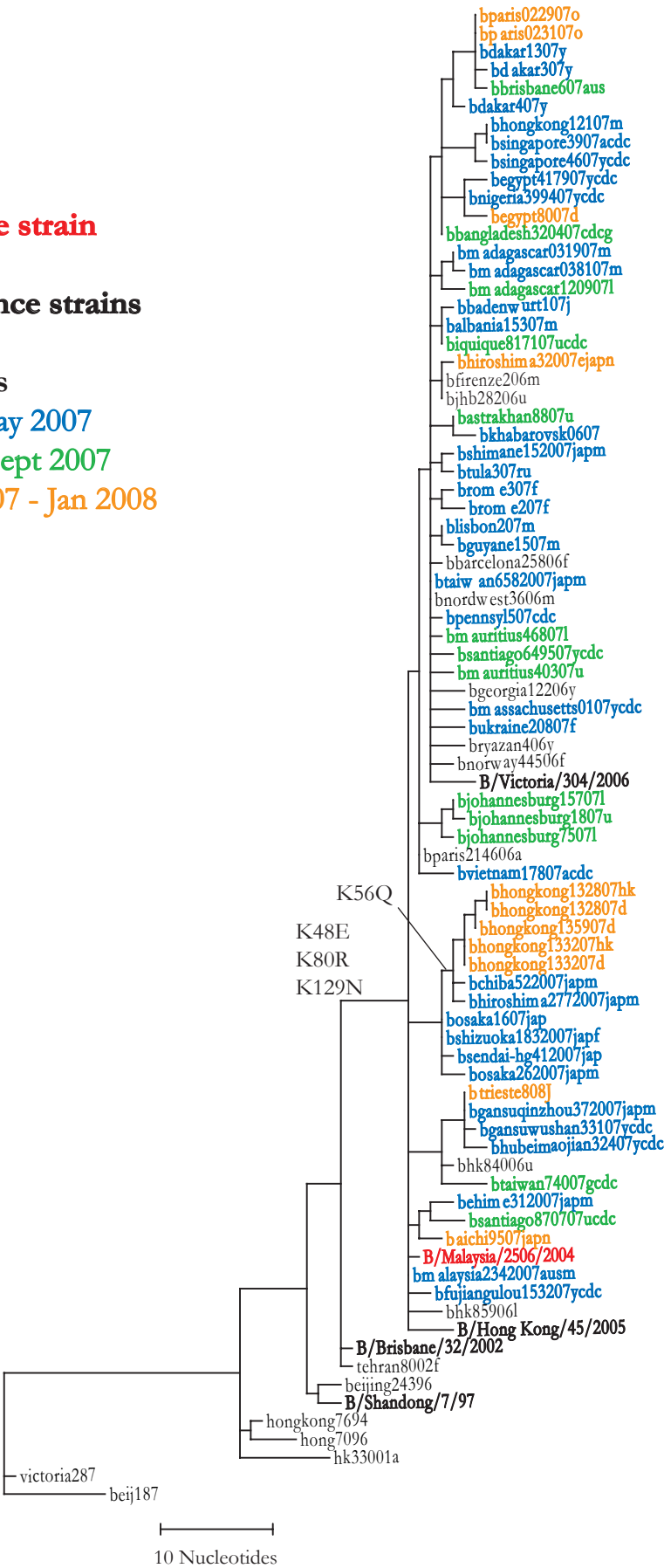


Figure 15. HA1 sequences of B viruses (Victoria lineage)

	1									100
B/Shandong/7/97					k				k	
B/Brisbane/32/2002					k				k	
B/Malaysia/2506/2004										
B/Hong Kong/45/2005										
B/Victoria/304/2006										
B/Rome/3/2007										
B/Ukraine/208/2007										
B/Lisbon/2/2007										
B/Mauritius/403/2007										
B/Johannesburg/157/2007										
B/Dakar/4/2007					e					
B/Pennsylvania/5/2007										
B/Khabarovsk/06/2007										
B/Paris/0229/2007										
B/Egypt/80/2007										
B/Hong Kong/1359/2007						q				
B/Madagascar/1209/2007										
B/Trieste/8/2008										
Consensus	DRICTGITSS	NSPHVVKTAT	QGEVNVTVGI	PLTTTPTKSH	FANLKGTETR	GKLCPKCLNC	TDL DVALGRP	KCTGNIP SAR	VSILHEVRPV	TSGCFPI MHD
	100		i	k						200
B/Shandong/7/97				k						k
B/Brisbane/32/2002				k						a
B/Malaysia/2506/2004	k									
B/Hong Kong/45/2005										a
B/Victoria/304/2006										t
B/Rome/3/2007										
B/Ukraine/208/2007									k	
B/Lisbon/2/2007										
B/Mauritius/403/2007										
B/Johannesburg/157/2007								p		
B/Dakar/4/2007										
B/Pennsylvania/5/2007										
B/Khabarovsk/06/2007										n
B/Paris/0229/2007							k			
B/Egypt/80/2007										n
B/Hong Kong/1359/2007										
B/Madagascar/1209/2007			t							
B/Trieste/8/2008										
Consensus	RTKIRQLPNL	LRGYEHIRLS	THNVINAENA	PGGPYKIGTS	GSCPNVTNGN	GFFATMAWAV	PKNDNNKTAT	NSLTIEVPYI	CTEGEDQITV	WGFHSDNETQ

Figure 15 continued

	200									300
B/Shandong/7/97										
B/Brisbane/32/2002										
B/Malaysia/2506/2004										
B/Hong Kong/45/2005										
B/Victoria/304/2006			d							
B/Rome/3/2007										
B/Ukraine/208/2007										
B/Lisbon/2/2007										
B/Mauritius/403/2007		v							f	
B/Johannesburg/157/2007										
B/Dakar/4/2007										
B/Pennsylvania/5/2007										
B/Khabarovsk/06/2007										
B/Paris/0229/2007										
B/Egypt/80/2007										
B/Hong Kong/1359/2007										
B/Madagascar/1209/2007									v	
B/Trieste/8/2008										
Consensus	MAKLYGDSKP	QKFTSSANGV	TTHYVSQIGG	FPNQTEDGGL	PQSGRIVVDY	MVQKSGKTGT	ITYQRGILLP	QKVWCASGRS	KVIKGSLLPLI	GEADCLHEKY
	300				347					
B/Shandong/7/97										
B/Brisbane/32/2002										
B/Malaysia/2506/2004										
B/Hong Kong/45/2005										
B/Victoria/304/2006										
B/Rome/3/2007										
B/Ukraine/208/2007										
B/Lisbon/2/2007										
B/Mauritius/403/2007										
B/Johannesburg/157/2007										
B/Dakar/4/2007										
B/Pennsylvania/5/2007										
B/Khabarovsk/06/2007										
B/Paris/0229/2007										
B/Egypt/80/2007										
B/Hong Kong/1359/2007										
B/Madagascar/1209/2007										
B/Trieste/8/2008										
Consensus	GGLNKSPPYY	TGEHAKAIGN	CPIWVKTPLK	LANGTKYRPP	AKLLKER					